Distributed LQR In this task the distributed discrete LQR of an LTI system is presented. The system is defined with

$$x(t+1) = Ax(t) + Bu(t) \tag{1}$$
$$y(t) = Cx(t) \tag{2}$$

Since the objective is to find the optimal input action that minimizes the cost function as

$$J(t) = x'_t Q x_t + \sum_{\tau=t}^{T-1} u_\tau^{\top} R u_\tau(t) \tag{3}$$

Note: $x_t^{\top}$ does not transpose. The cost function is a scalar, the transpose is the transpose, but it means A_t on the power T, which is the horizon, the number of steps the optimization optimizes for

$$x_{t,T} = A_{t,\tau} x_\tau + B_{t,\tau,\tau-1} \tag{4}$$

where

$$x_{t,T} = \begin{pmatrix} x_t(t) \\ \vdots \\ x_T(t) \end{pmatrix} \quad u_{\tau,\tau-1} = \begin{pmatrix} u_\tau(0) \\ \vdots \\ u_\tau(T-1) \end{pmatrix} \tag{5}$$

and

$$A_t = \begin{pmatrix} A_t \\ \vdots \\ A_T \end{pmatrix} \quad B_t = \begin{pmatrix} B_t & & \\ \vdots & \ddots & \\ B_T & \cdots & B_T \end{pmatrix} \tag{6}$$

coupling constraint

$$x_t(T) = x_T(t) \tag{7}$$

decoupled constraint

$$|u_\tau(t)| \leq u_{\max} \forall t \in t \, \forall \tau \, T \tag{8}$$

Loss function, since the first state is given, so it is unaffected by the control input, it is not included in the computation.

$$P_{t,\tau_{\max},\tau-1,\tau_{\max}-1} = \sum_{\tau=t}^{\tau_{\max}-1} x_\tau^{\top}(t) Q x_\tau(t) + u_\tau^{\top}(t) R u_\tau(t) \tag{9}$$
$$P_{t,\tau_{\max},\tau-1,\tau_{\max}-1} = \sum_{\tau=t}^{\tau_{\max}-1} \begin{pmatrix} x_\tau \\ u_\tau \end{pmatrix}^{\top} \begin{pmatrix} Q & 0 \\ 0 & R \end{pmatrix} \begin{pmatrix} x_\tau \\ u_\tau \end{pmatrix} \tag{10}$$

The coupling constraint can be expressed as

$$V^{(\tau-1)}(x_{t,\tau_{\max}}(\tau-1), x_{t,\tau_{\max}}(\tau-1)) = \tag{11}$$
with
$$x_{t,\tau_{\max}}(\tau-1) = A_{t,\tau_{\max}} x_{t,\tau_{\max}}(\tau-1) + B_{t,\tau_{\max},\tau-1} \tag{12}$$
$$x_{t,\tau_{\max}}(\tau_{\max}-1) = A_{t,\tau_{\max}} x_{t,\tau_{\max}}(\tau_{\max}-1) + B_{t,\tau_{\max},\tau_{\max}-1} \tag{13}$$
where
$$A = (A^{\top} \cdots A_t \quad \cdots \quad A_t)^{\top} \quad B_t \quad (A_t^{\top} \cdots A_t)^{\top} \tag{14}$$

therefore the Lagrangian of problem is formulated as

$$\mathcal{L}_{t,\tau_{\max}} = \sum_{\tau=t}^{\tau_{\max}-1} x_\tau^{\top}(t) Q x_\tau(t) + u_\tau^{\top}(t) R u_\tau(t) + \lambda_\tau^{\top}(t)(x_\tau(t) - B_{t,\tau,\tau-1} u_{\tau,\tau-1}) \tag{15}$$
$$+ \sum_{\tau=t}^{\tau_{\max}-1} \mu^{(\tau)\top}(\lambda_\tau^{\top} x_\tau + B_{t,\tau,\tau-1} u_{\tau,\tau-1} - \lambda_{\tau+1}^{\top}(x_{\tau+1} - B_{t,\tau+1,\tau} u_{\tau+1})) \tag{16}$$
s.t.
$$|u_\tau(t)| \leq u_{\max} \forall t \in t \, \forall \tau \, T \tag{17}$$

The dual function is separated into parts that only depend on one input sequence $u_{\tau,\tau-1}$. Equation 20. This allows individual optimization for each input sequence. The dual function Equation 21

$$d(\lambda) = \inf_{u_{\tau,\tau-1}} \{\mathcal{L}(x_{t,\tau_{\max}}(\tau-1)^{\top} x_{t,\tau_{\max}}(\tau-1), u_{\tau,\tau-1}) - \lambda^{\top} x_{t,\tau_{\max}}(\tau-1)\} \tag{18}$$
$$d_1(\lambda^{(\tau)}, \lambda^{(\tau)}) = \inf_{u_{\tau,\tau-1}} \{\mathcal{L}(x_{t,\tau_{\max}}(\tau-1)^{\top} x_{t,\tau_{\max}}(\tau-1), u_{\tau,\tau-1}) - \lambda^{(\tau)\top} x_{t,\tau_{\max}}(\tau-1)\} \tag{19}$$
$$d_2(\lambda^{(\tau)}, \lambda^{(\tau)}) = \inf_{u_{\tau,\tau-1}} \{\mathcal{L}(x_{t,\tau_{\max}}(\tau-1)^{\top} x_{t,\tau_{\max}}(\tau-1), u_{\tau,\tau-1}) - \lambda^{(\tau)\top} x_{t,\tau_{\max}}(\tau-1)\} \tag{20}$$
s.t.
$$|u_\tau(t)| \leq u_{\max} \forall t \in t \, \forall \tau \, T \tag{21}$$

Since this optimization problem includes information that is computed with respect to λ_τ, opposing the explicit formulation, can't be solved in one go. The optimal input value need to be updated which is done in iterations, therefore it is updated at a recipient decent step with step size α **SC42125Lecture6**.

$$\lambda_{\tau+1} = \lambda_\tau + \alpha \nabla_{\lambda_\tau} d_1(\lambda_\tau) \tag{22}$$

Given the dynamics of the plants are independent of the input action , and they differ at their local state Equation. the process results in the process that the iteration of the terminal state is plotted at Figure 1. The process is validated by comparing the results to the centralized optimization. Both methods produce the same result of Equation 34

$$\lambda_{\tau+1} = \arg\min_{\lambda} \sum_{\tau=t}^{\tau_{\max}} d_1(\lambda_\tau^{(\tau)}, \lambda_\tau^{(\tau)}) \tag{23}$$

where the quadratic loss part λ of the dual function does not depend on it therefore it can be written as

$$\lambda_{\tau+1} = \arg\min_{\lambda} \sum_{\tau=t}^{\tau_{\max}} \lambda_\tau^{\top} d_1(\lambda_\tau^{(\tau)}, \lambda_\tau^{(\tau)}) \quad \forall \tau \, t \tag{24}$$

$$\lambda_{\tau+1} = \arg\min_{\lambda} \sum_{\tau=t}^{\tau_{\max}} d_1(\lambda_\tau^{(\tau)}, \lambda_\tau^{(\tau)}) \tag{25}$$

$$\begin{aligned} \lambda_1 &= \begin{bmatrix} 0.6246 & -0.0035 \\ 0.0719 & 0.828 \end{bmatrix} \quad \lambda_2 = \begin{bmatrix} -0.07 & 0 \\ 0.1744 & 1 \end{bmatrix} \quad \forall \tau \, t \\ \lambda_{\tau_1} &= \begin{bmatrix} -0.6 \\ 0.1 \end{bmatrix} \quad \lambda_{\tau_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \lambda_{\tau_3} = \begin{bmatrix} 0.07 \\ 0.1744 \end{bmatrix} \quad \lambda_{\tau_4} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned} \tag{26}$$

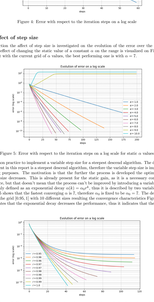

Figure 1: Base Case Distributed LQR with $\alpha = 5$

$$x_{t,T} = \begin{pmatrix} -0.0035 \\ 0.8288 \end{pmatrix} \forall t \, T \tag{27}$$

This solution results in the evolution of the state Figure 1 which is compared to centralized solution is almost identical with an mean square difference of only 3.8e - 11

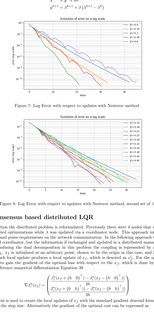

Figure 2: Base Case Distributed LQR evolution with $\alpha = 5$

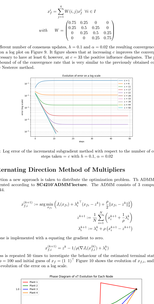

Figure 3: Centralized LQR evolution

In order to show that the distributed process converges the error is plotted with respect to the iteration step Equation 35. The error is defined as the difference between the state and the centralized solution. Given if this process is zero than the finality gap goes to zero, which proves strong duality and then validates the process. The error is calculated by taking the sum of the absolute value of the error at each iteration step. Given if the error goes to zero at a specific step, the optimal input action is with $\alpha = 7$.

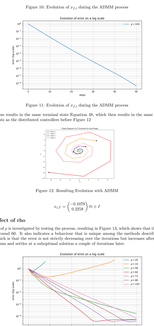

Figure 4: Error with respect to the iteration steps on a log scale

0.1 Effect of step size

In this section the effect of step size is investigated on the evolution of the error over the iterations. First, the effect of changing the state value of a constant α on the range is simulated on Figure 5. It shows that with the current grid of α values, the best performance is with $\alpha = 7$.

Figure 5: Error with respect to the iteration steps on a log scale for static α values

It is common practice to implement a variable step size for a steeper decent algorithm. The distributed LQR present in this report is a steepest decent algorithm, therefore a variable step size is implemented for testing purposes. The motivation is that the further the process is developed the optimal update approaches zero, therefore the step size should be reduced. The step size is reduced by multiplying by a constant, but doesn't mean that the process can't be improved by introducing a variable α. $\alpha(t)$ is commonly defined as an exponential decay $\alpha(t) = \alpha_0 e^{\lambda t}$, thus it is described by two variables α_0 and λ. Figure 6 shows the effect of different initial step sizes with a constant $\lambda = 0.1$. Figure 7 shows the effect of λ tested on the grid λ, with 10 different values resulting the convergence characteristic Figure 8. The plot indicates that the exponential decay decreases the performance, thus it indicates that the ideal α is static.

Figure 6: Error with respect to the iteration steps on a log scale for dynamic α values

1 Nesterov method

The Nesterov method is implemented according to **nesterov100pointhard**, which introduces a momentum term to improve the convergence. The term β is used to control the momentum, which accelerates the optimum. It produces improvements over $\beta \in [0.1, 0.6]$ Figure 7. To further investigate the behaviour, this segment is plotted on Figure 8. The figure shows that for a maximum value of $\beta = 0.28$ the convergence happens at an exponential rate, while this breaks for $\beta \geq 0.32$, therefore the optimal value for β is chosen to be 0.28. Then the convergence increased is rate from reaching the stopping condition from 42 steps to 22 steps, which is a 45% increase in performance.

Figure 7: Log Error with respect to updates with Nesterov method

Figure 8: Log Error with respect to updates with Nesterov method, second set of β

2 Consensus based distributed LQR

In this section the distributed problem is reformulated. Previously there were 2 nodes that were doing the separated optimization while in the upcoming section there is a coordinator node. This approach introduces new node and poses requirements on the network communication. In the following approach there is no centralized coordinator, but the information is exchanged between the nodes in a distributed manner. Instead confusing the dual decomposition in this problem the coupling is represented by a coupling matrix $\mathcal{L}$ and $\mathcal{L}_{\tau}$ is the τ row of the coupling matrix. The goal is to find the $\mathcal{L}$ that is shared locally. Each local update produces a local update of μ, which is denoted as $\mu_\tau^{(\tau)}$. For the update it is necessary to gain the gradient of the optimal loss with respect to the μ_τ which is done by using the central difference numerical differentiation Equation 36

$$\nabla \mathcal{L}(\mu_\tau) = \begin{pmatrix} \frac{\mathcal{L}(\mu_\tau + (h \quad 0)^{\top}) - \mathcal{L}(\mu_\tau - (h \quad 0)^{\top})}{2h} \\ \frac{\mathcal{L}(\mu_\tau + (0 \quad h)^{\top}) - \mathcal{L}(\mu_\tau - (0 \quad h)^{\top})}{2h} \end{pmatrix} \tag{28}$$

The gradient is used to create the local update of μ_τ with the standard gradient decent formula, where α is again the step size. Alternatively the gradient of the optimal cost can be expressed as

$$\nabla \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mu_\tau^{(\tau)}} \frac{\partial \mu_\tau^{(\tau)}}{\partial \mu_\tau} \tag{29}$$
$$= 2(\mathcal{L}^{\top} (A_\tau + B_{\tau,\tau,\tau-1} \mathcal{L}_{\tau,\tau-1}) \, \mathcal{L}_{\tau,\tau-1}) (\mathcal{L}^{\top} B)^{\top} B^{\top} \tag{30}$$

where $\mathcal{L}_{\tau,\tau-1}$ is computed as the optimal input sequence as defined before at $\mu_\tau^{(\tau)}$. This alternative also offers the reduction of optimization to be solved from 4 to 1.

$$\mu_\tau^{(\tau)} = \mu_\tau - \alpha \nabla \mathcal{L}(\mu_\tau) \tag{31}$$

Following the local update each plant for every node the results are coupled by the consensus update. In the consensus update the μ_τ values are averaged with respect to a symmetric doubly stochastic matrix W **SC42125Lecture7**. The consensus update can is repeated a parameter ν amount of times.

$$\nu 1 = \sum_{\tau=1}^{N} W(\tau, j) \mu_\tau^{(\tau)} \quad \forall \tau \, t \tag{32}$$
$$\text{with} \quad W = \begin{pmatrix} 0.75 & 0.25 & 0 & 0 \\ 0.25 & 0.5 & 0.25 & 0 \\ 0 & 0.25 & 0.5 & 0.25 \\ 0 & 0 & 0.25 & 0.75 \end{pmatrix} \tag{33}$$

For the different number of consensus updates, $\nu = 0$ and $\nu = 0.05$ the resulting convergence rates are displayed on a log plot on Figure 9. It figure shows that at increasing ν improves the convergence, and is even necessary to have at least 5 iterations to see a fit the positive tolerance diagram. The plot shows an upper bound of the convergence rate that is very similar to the previously obtained convergence rate of the Nesterov method.

Figure 9: Log error of the incremental subgradient method with respect to the number of consensus steps taken $\nu = 0$ with $h = 0.1$, $\alpha = 0.02$

3 Alternating Direction Method of Multipliers

In this section a new approach is taken to distribute the optimization problem. The ADMM approach is implemented according to **SC42125 Lecture8**. The ADMM consists of 3 computations as Equation 44.

$$\mu_\tau^{(\tau)+1} = \arg\min_{\mu} \{L(\mu_\tau) + \lambda^{\top} (x_\tau - \beta^{\top}) + \frac{\rho}{2} \|x_\tau - \beta^{\top}\|\} \tag{34}$$
$$\beta^{\tau+1} = \sum_{\tau=1}^{N} \left(\lambda^{(\tau)} + x_\tau^{\top} \right) \tag{35}$$
$$\lambda_\tau^{\tau+1} = \lambda_\tau^{\tau} + \alpha(\beta^{\tau+1} - \beta^{\tau+1}) \tag{36}$$

The first one is implemented with a repeating the gradient in step.

$$x_\tau^{(\tau)+1} = x_\tau^{\tau} - \xi(\rho \nabla \mathcal{L}(x_\tau^{\tau}) + \lambda_\tau^{\tau}) \tag{37}$$

The process is repeated 50 times to investigate the behaviour of the estimated terminal states of each node. At $\rho = 100$ and initial guess of $x_\tau = (1 \quad 1)^{\top}$ Figure 10 shows the evolution of $x_{\tau,t}$, and Figure 11 shows the evolution of the error on a log scale.

Figure 10: Evolution of $x_{\tau,t}$ during the ADMM process

Figure 11: Evolution of $x_{\tau,t}$ during the ADMM process

This process results in the same terminal state Equation 38, which then results in the same behaviour of the plants as the distributed consensus Figure 12.

$$x_{t,T} = \begin{pmatrix} -0.0035 \\ 0.8288 \end{pmatrix} \forall t \, T \tag{38}$$

Figure 12: Bounding Evolution with ADMM

3.1 Effect of ρ

The ρ is investigated by testing the process, resulting in Figure 13, which shows that the optimal value is around 80. It also indicates a behaviour that is unique among the methods described in this report, which is that the error is not strictly decreasing over the iterations but increases after attaining the minimum and settles at a suboptimal solution a couple of iterations later.

Figure 13: ADMM evolution of error with respect to ρ